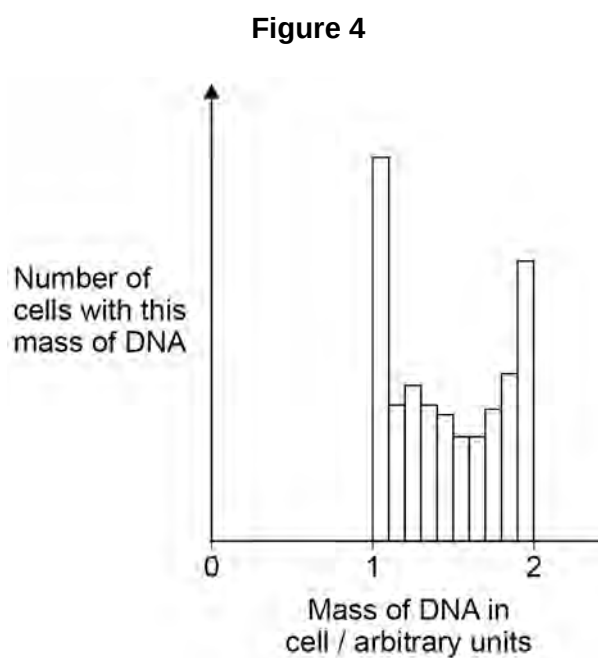


07.1

Figure 4 shows the mass of DNA present in a group of healthy cells.

Do not write
outside the
box



Use your knowledge of the cell cycle to explain the results shown in **Figure 4**.

[3 marks]



0 7 . 2

Suggest **one** way **Figure 4** would be different if these cells were tumour cells.

Justify your answer.

[2 marks]

0 7 . 3

Describe the behaviour of chromosomes in prophase and metaphase of mitosis.

[2 marks]

Prophase _____

Metaphase _____

Question 7 continues on the next page

Turn over ►



07.4

During anaphase, the spindle exerts 3×10^{-11} N of force on each chromatid. This force generates 6×10^{-19} W of power.

Calculate the speed of movement, in nm s^{-1} , of one chromatid during anaphase using the following equation:

$$P = F \times V$$

Where P = power in W

F = force in N

V = speed in m s^{-1}

Show your working.

[2 marks]

Answer _____ nm s^{-1}

9

